

GEARS Human Validation — Reviewer 1 (Blinded)

Purpose

Independently determine how each citing paper uses GEARS. Machine labels, confidence, agent outputs, and machine-selected evidence are intentionally hidden.

Workflow

Complete every row in the displayed order. Do not open the other reviewer document or the machine-selection file until both independent reviews are locked.

Use only the linked approved article versions. Inspect every GEARS occurrence and its surrounding methods, results, figures, tables, and available supplements.

After both reviewers lock their answers, discuss only disagreements in a separate consensus record.

Allowed values

Review status: DRAFT / LOCKED

Citation validity and method executed: Yes / No / Unclear

Uses (select all): Mention only / Benchmark / Extend or develop / Apply to biological data / Other executed use / Unclear

Primary use: Mention only / Benchmark / Extend or develop / Apply to biological data / Other / Unclear

Biological insight: Yes / No / Unclear

Evidence sufficiency: Sufficient / Incomplete / Citation not located / Unclear

Supporting evidence: shortest exact quote plus section, page, paragraph, figure, or table locator.

Review table

Record	Article and approved evidence links	Reviewer 1 independent record
GEARS-H V-023 FORMAL VALIDATI ON	GPerturb: Gaussian process modelling of single-cell perturbation data DOI: 10.1038/s41467-025-61165-7 Article page: https://doi.org/10.1038/s41467-025-61165-7 Full text 1: https://www.nature.com/articles/s41467-025-61165-7.pdf Controller-registered target occurrence locator(s): occ-0001 (document_650988dd5720f43d70f3.txt:p-0002); occ-0002 (document_650988dd5720f43d70f3.txt:p-0002); occ-0003 (document_650988dd5720f43d70f3.txt:p-0002); occ-0004 (document_650988dd5720f43d70f3.txt:p-0002); occ-0005 (document_650988dd5720f43d70f3.txt:p-0002); occ-0006 (document_650988dd5720f43d70f3.txt:p-0002); occ-0007 (document_650988dd5720f43d70f3.txt:p-0003); occ-0008 (document_650988dd5720f43d70f3.txt:p-0003); occ-0009 (document_650988dd5720f43d70f3.txt:p-0003); occ-0010 (document_650988dd5720f43d70f3.txt:p-0004); occ-0011 (document_650988dd5720f43d70f3.txt:p-0004); occ-0012 (document_650988dd5720f43d70f3.txt:p-0004); occ-0013	Review status:LOCKED Citation validity:Yes Method executed:Yes Uses (select all): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: We first compared the predictive

	(document_650988dd5720f43d70f3.txt:p-0004); occ-0014 (document_650988dd5720f43d70f3.txt:p-0004); occ-0015 (document_650988dd5720f43d70f3.txt:p-0004); occ-0016 (document_650988dd5720f43d70f3.txt:p-0004); occ-0017 (document_650988dd5720f43d70f3.txt:p-0004); occ-0018 (document_650988dd5720f43d70f3.txt:p-0004); occ-0019 (document_650988dd5720f43d70f3.txt:p-0004); occ-0020 (document_650988dd5720f43d70f3.txt:p-0004); occ-0021 (document_650988dd5720f43d70f3.txt:p-0004); occ-0022 (document_650988dd5720f43d70f3.txt:p-0005); occ-0023 (document_650988dd5720f43d70f3.txt:p-0005); occ-0024 (document_650988dd5720f43d70f3.txt:p-0005); occ-0025 (document_650988dd5720f43d70f3.txt:p-0005); occ-0026 (document_650988dd5720f43d70f3.txt:p-0005); occ-0027 (document_650988dd5720f43d70f3.txt:p-0005); occ-0028 (document_650988dd5720f43d70f3.txt:p-0005); occ-0029 (document_650988dd5720f43d70f3.txt:p-0007); occ-0030 (document_650988dd5720f43d70f3.txt:p-0007); occ-0031 (document_650988dd5720f43d70f3.txt:p-0012); occ-0032 (document_650988dd5720f43d70f3.txt:p-0012); occ-0033 (document_650988dd5720f43d70f3.txt:p-0012); occ-0034 (document_650988dd5720f43d70f3.txt:p-0012); occ-0035 (document_650988dd5720f43d70f3.txt:p-0013)	performance of GPerturb, CPA, GEARS and SAMS-VAE on a subset of the genome-wide CRISPR interference Perturb-seq dataset. Notes:
GEARS-H V-046 FORMAL VALIDATI ON	Generative genomics accurately predicts future experimental results DOI: 10.1101/2025.09.08.674753 Article page: https://doi.org/10.1101/2025.09.08.674753 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/09/09/2025.09.08.674753.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_982567858905463cf522.txt:p-0093)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Despite being a valuable resource for AI model development^{20,34,35}, Perturb-seq data presents unique challenges, including technical artifacts and non-specific responses^{20,36–39}, which require focused resolution for this data type that goes beyond the

		scope of our initial study. Notes:
GEARS-H V-024 FORMAL VALIDATION	scPRINT: pre-training on 50 million cells allows robust gene network predictions DOI: 10.1038/s41467-025-58699-1 Article page: https://doi.org/10.1038/s41467-025-58699-1 Full text 1: https://www.nature.com/articles/s41467-025-58699-1.pdf Controller-registered target occurrence locator(s): occ-0001 (document_59b9ce377d6f923515c3.txt:p-0020)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Understanding the cellular mechanism is considered a milestone in biology, allowing us to predict cell behavior and the impact of drugs and gene knock-outs1,2,3,4,5. Notes:
GEARS-H V-027 FORMAL VALIDATION	Prediction of cellular morphology change under perturbations with transcriptome-guided diffusion model DOI: 10.1101/2025.07.16.665106 Article page: https://doi.org/10.1101/2025.07.16.665106 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/07/16/2025.07.16.665106.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_4606fd53ac6e26c3d47a.txt:p-0021); occ-0002 (document_4606fd53ac6e26c3d47a.txt:p-0031)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No

		Evidence sufficiency:Sufficient Supporting quote + locator: Therefore, integrating MorphDiff with methods that can predict changes in gene expression under unseen perturbations, such as GEARS47, is a possible extension. Notes:
GEARS-H V-028 FORMAL VALIDATION	Causal models and prediction in cell line perturbation experiments DOI: 10.1186/s12859-024-06027-7 Article page: https://doi.org/10.1186/s12859-024-06027-7 Full text 1: https://www.researchsquare.com/article/rs-3973384/latest.pdf Full text 2: https://bmcbioinformatics.biomedcentral.com/counter/pdf/10.1186/s12859-024-06027-7 Controller-registered target occurrence locator(s): occ-0001 (document_8e546d952becc9f38df6.txt:p-0018); occ-0002 (document_d18339b76d624d106839.txt:p-0005); occ-0003 (document_d18339b76d624d106839.txt:p-0023)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: [10] developed GEARS to predict the effect of double knockouts/knockdowns using in vitro responses to single knockouts/knockdowns. Notes:

GEARS-H V-029 FORMAL VALIDATION	HisToSpatialCNV: an interpretable deep learning method predicting spatial copy number variations from histopathology images DOI: 10.1038/s41551-026-01754-z Article page: https://doi.org/10.1038/s41551-026-01754-z Full text 1: https://www.nature.com/articles/s41551-026-01754-z.pdf Controller-registered target occurrence locator(s): occ-0001 (document_1f68275d2a59b48d816a.txt:p-0018)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Similar observations were also shared by others, in that single-cell FMs currently do not necessarily outperform established task-specific methods across all downstream tasks^{74,75,76,77}. Notes:
GEARS-H V-006 FORMAL VALIDATION	Disentanglement of single-cell data with biolord DOI: 10.1038/s41587-023-02079-x Article page: https://doi.org/10.1038/s41587-023-02079-x Full text 1: https://www.nature.com/articles/s41587-023-02079-x.pdf Controller-registered target occurrence locator(s): occ-0001 (document_1075fecb1136e9480c16.txt:p-0003); occ-0002 (document_1075fecb1136e9480c16.txt:p-0003); occ-0003 (document_1075fecb1136e9480c16.txt:p-0003); occ-0004 (document_1075fecb1136e9480c16.txt:p-0003); occ-0005 (document_1075fecb1136e9480c16.txt:p-0003); occ-0006 (document_1075fecb1136e9480c16.txt:p-0003); occ-0007 (document_1075fecb1136e9480c16.txt:p-0005); occ-0008 (document_1075fecb1136e9480c16.txt:p-0009); occ-0009 (document_1075fecb1136e9480c16.txt:p-0009); occ-0010	Review status:LOCKED Citation validity:Yes Method executed:Yes Uses (select all): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: We show that

	(document_1075fecb1136e9480c16.txt:p-0009); occ-0011 (document_1075fecb1136e9480c16.txt:p-0009); occ-0012 (document_1075fecb1136e9480c16.txt:p-0009); occ-0013 (document_1075fecb1136e9480c16.txt:p-0009); occ-0014 (document_1075fecb1136e9480c16.txt:p-0009); occ-0015 (document_1075fecb1136e9480c16.txt:p-0012); occ-0016 (document_1075fecb1136e9480c16.txt:p-0012); occ-0017 (document_1075fecb1136e9480c16.txt:p-0012)	biolord outperforms GEARs in the prediction of unseen one-gene perturbation (normalized mean squared error, one of one gene unseen; GEARs: 0.47; biolord: 0.37) and two-gene perturbations (normalized mean squared error, two of two genes unseen; GEARs: 0.53; biolord: 0.50, one of one gene unseen; GEARs: 0.39; biolord: 0.35, zero of two genes unseen; GEARs: 0.28; biolord: 0.20; Methods; Supplementary Note 3 and Extended Data Fig. 2). Notes:
GEARS-H V-043 FORMAL VALIDATION	Predicting Multigene Transcriptional Outcomes Based on Dual-Stream Mamba Neural Network DOI: 10.12677/biphy.2025.131001 Article page: https://doi.org/10.12677/biphy.2025.131001 Full text 1: https://drpress.org/ojs/index.php/fcis/article/download/31264/30614 Full text 2: https://pdf.hanspub.org/biphy_2790105.pdf Controller-registered target occurrence locator(s): occ-0001 (document_6ff398dc26411036850e.txt:p-0004); occ-0002 (document_6ff398dc26411036850e.txt:p-0005); occ-0003 (document_6ff398dc26411036850e.txt:p-0005); occ-0004 (document_6ff398dc26411036850e.txt:p-0008); occ-0005 (document_6ff398dc26411036850e.txt:p-0024); occ-0006 (document_fbe3d2050efb71b5d261.txt:p-0018); occ-0007 (document_fbe3d2050efb71b5d261.txt:p-0018); occ-0008	Review status:LOCKED Citation validity:Yes Method executed:Yes Uses (select all): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: The experimental results in Figure 4

	(document_fbe3d2050efb71b5d261.txt:p-0023); occ-0009 (document_fbe3d2050efb71b5d261.txt:p-0051)	demonstrate that the proposed model exhibits significant advantages across all data groups, with prediction performance substantially surpassing all baseline models Notes:
GEARS-H V-047 FORMAL VALIDATION	Prediction of cellular morphology changes under perturbations with a transcriptome-guided diffusion model DOI: 10.1038/s41467-025-63478-z Article page: https://doi.org/10.1038/s41467-025-63478-z Full text 1: https://www.nature.com/articles/s41467-025-63478-z.pdf Controller-registered target occurrence locator(s): occ-0001 (document_39a6578d531a744400b1.txt:p-0011); occ-0002 (document_39a6578d531a744400b1.txt:p-0017)	Review status: Citation validity: Method executed: Uses (select all): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:
GEARS-H V-014 FORMAL VALIDATION	scMomer: A modality-aware pretraining framework for single-cell multi-omics modeling under missing modality conditions DOI: 10.1101/2025.08.04.668374 Article page: https://doi.org/10.1101/2025.08.04.668374 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/08/05/2025.08.04.668374.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_32bff1ac12fad55c6696.txt:p-0018); occ-0002 (document_32bff1ac12fad55c6696.txt:p-0018); occ-0003 (document_32bff1ac12fad55c6696.txt:p-0025); occ-0004 (document_32bff1ac12fad55c6696.txt:p-0026); occ-0005	Review status:LOCKED Citation validity:Yes Method executed:Yes Uses (select all): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient

	(document_32bff1ac12fad55c6696.txt:p-0026); occ-0006 (document_32bff1ac12fad55c6696.txt:p-0029)	Supporting quote + locator: Following the setup used in CellFM and scFoundation, we integrated gene embeddings from scMomer into the GEARS13 for downstream prediction (Supplementary Fig. 11). Notes:
GEARS-H V-008 FORMAL VALIDATION	scFormer: A Universal Representation Learning Approach for Single-Cell Data Using Transformers DOI: 10.1101/2022.11.20.517285 Article page: https://doi.org/10.1101/2022.11.20.517285 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2022/11/22/2022.11.20.517285.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_c8dd8f4837f2553f5226.txt:p-0009); occ-0002 (document_c8dd8f4837f2553f5226.txt:p-0009); occ-0003 (document_c8dd8f4837f2553f5226.txt:p-0011)	Review status: LOCKED Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: We compare the performance against the recent GEARS method (Roohani et al., 2022) and the multi layer perceptron baseline. Notes:
GEARS-H V-002	Single-cell technologies: current and near future DOI: 10.1007/s11427-024-2813-y Article page: https://doi.org/10.1007/s11427-024-2813-y	Review status: LOCKED Citation validity: Yes

FORMAL VALIDATI ON	Full text 1: https://link.springer.com/content/pdf/10.1007/s11427-024-2813-y.pdf Controller-registered target occurrence locator(s): occ-0001 (document_ae8274926b66104044e8.txt:p-0004)	Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: For scCRISPR data analysis, existing bioinformatics tools generally focus on three aspects, including data preprocessing (Wei et al., 2023), data denoising (Duan et al., 2019), and functional analysis (Roohani et al., 2024), aiming to uncover rich perturbation information. Notes:
GEARS-H V-049 FORMAL VALIDATI ON	Decoding cell identity with multi-scale explainable deep learning DOI: 10.1101/2024.02.05.578922 Article page: https://doi.org/10.1101/2024.02.05.578922 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2024/02/08/2024.02.05.578922.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_c52b63babff855635323.txt:p-0054)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient

		Supporting quote + locator: Cell Decoder opens up the possibility of predicting cell responses to genetic perturbations⁵⁷. Notes:
GEARS-H V-004 FORMAL VALIDATION	Multimodal Benchmarking of Foundation Model Representations for Cellular Perturbation Response Prediction DOI: 10.1101/2025.06.26.661186 Article page: https://doi.org/10.1101/2025.06.26.661186 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/06/28/2025.06.26.661186.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_4b239dec33d56e05c63a.txt:p-0006)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: There are many specialized perturbation response models one could use on top of such embeddings (e.g. Roohani et al. (2024), Lotfollahi et al. (2023), Piran et al. (2024)), but those models are beyond the scope of our work, which intentionally fixes a single model with a consistent training protocol. Notes:

GEARS-H V-031 FORMAL VALIDATI ON	Simple Causal Relationships in Gene Expression Discovered through Deep Learned Collective Variables DOI: 10.1101/2023.01.18.524617 Article page: https://doi.org/10.1101/2023.01.18.524617 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/01/20/2023.01.18.524617.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_c4e884dd4ef14a20ead1.txt:p-0060)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: The use of autoencoders [9], UMAP [10,11], and other methods of dimensionality reduction [12,13] have demonstrated that it is possible to discover relationships in statistical covariation of many genes that can be used, for example, to distinguish different cell types and lineages [14], to predict mean gene expression under perturbation [15,16], to predict the interactions between genes under combinatorial perturbations [17], and to generate counterfactual predictions on single-cell response under new drug treatments [18] or novel perturbations [19].
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		autoencoder, together with GEARs4 and graphVCI5 graph neural network models disentangle perturbation covariates with basal cellular states, thus providing more flexibility for the in silico perturbation analysis. Notes:
GEARS-H V-017 FORMAL VALIDATION	A General Single-Cell Analysis Framework via Conditional Diffusion Generative Models DOI: 10.1101/2023.10.13.562243 Article page: https://doi.org/10.1101/2023.10.13.562243 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/10/17/2023.10.13.562243.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_fa1b8e373828243f0f02.txt:p-0001); occ-0002 (document_fa1b8e373828243f0f02.txt:p-0013); occ-0003 (document_fa1b8e373828243f0f02.txt:p-0013); occ-0004 (document_fa1b8e373828243f0f02.txt:p-0018); occ-0005 (document_fa1b8e373828243f0f02.txt:p-0018); occ-0006 (document_fa1b8e373828243f0f02.txt:p-0018); occ-0007 (document_fa1b8e373828243f0f02.txt:p-0018); occ-0008 (document_fa1b8e373828243f0f02.txt:p-0018); occ-0009 (document_fa1b8e373828243f0f02.txt:p-0018); occ-0010 (document_fa1b8e373828243f0f02.txt:p-0018); occ-0011 (document_fa1b8e373828243f0f02.txt:p-0018); occ-0012 (document_fa1b8e373828243f0f02.txt:p-0022); occ-0013 (document_fa1b8e373828243f0f02.txt:p-0039); occ-0014 (document_fa1b8e373828243f0f02.txt:p-0039); occ-0015 (document_fa1b8e373828243f0f02.txt:p-0039); occ-0016 (document_fa1b8e373828243f0f02.txt:p-0039)	Review status:LOCKED Citation validity:Yes Method executed:Yes Uses (select all): Primary use:Extend/Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Under such zero-shot setting, we incorporate biological prior for the unseen genes by adapting the gene ontology-based graph neural network from GEARs, as detailed in Section 2.2 Notes:

GEARS-H V-044 FORMAL VALIDATION	BIOFORMERS: A SCALABLE FRAMEWORK FOR EXPLORING BIOSTATES USING TRANSFORMERS DOI: 10.1101/2023.11.29.569320 Article page: https://doi.org/10.1101/2023.11.29.569320 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/12/01/2023.11.29.569320.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_c16d75fa50a80feb1a20.txt:p-0013)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: In the one gene perturbation experiments, we adhered to the preprocessing protocol established by Roohani et al. [26], which includes normalization by total molecular counts, log transformation of the data, selection of 5,000 highly variable molecules, and inclusion of any perturbed molecules otherwise omitted. Notes:
GEARS-H V-050 FORMAL VALIDATION	Perturbation Atlases as Pharmacological Reasoning Engines for Connecting Molecular Interventions to Context-Dependent Cellular Responses DOI: 10.51847/a1xh4xuw5o Article page: https://doi.org/10.51847/a1xh4xuw5o Full text 1: https://pharmacophorejournal.com/storage/files/article/aa68d254-a799-4a9c-8a30-0f7cf119a88a-Y08w047JPZKDeQ4r/9rBBC2PqJpy90ji.pdf	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No

	Controller-registered target occurrence locator(s): occ-0001 (document_b1433a11e3f2f5132e34.txt:p-0015); occ-0002 (document_b1433a11e3f2f5132e34.txt:p-0022)	Evidence sufficiency:Sufficient Supporting quote + locator: GEARS shows how graph-based information can contribute to prediction of transcriptional outcomes for novel multigene perturbations [16]. Notes:
GEARS-H V-045 FORMAL VALIDATION	Primer on machine learning applications in brain immunology DOI: 10.3389/fbinf.2025.1554010 Article page: https://doi.org/10.3389/fbinf.2025.1554010 Full text 1: https://www.frontiersin.org/journals/bioinformatics/articles/10.3389/fbinf.2025.1554010/pdf Controller-registered target occurrence locator(s): occ-0001 (document_ec05b2c2a5b977b9f169.txt:p-0007)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: For example, machine learning tools can be used to enhance dimensionality reduction and integration of large datasets, but also for more sophisticated tasks, such as predictive modeling of gene perturbations (Xu et al., 2021; Lotfollahi et al., 2022; Roohani et al., 2024).

		Notes:
GEARS-H V-016 FORMAL VALIDATION	Systema: a framework for evaluating genetic perturbation response prediction beyond systematic variation DOI: 10.1038/s41587-025-02777-8 Article page: https://doi.org/10.1038/s41587-025-02777-8 Full text 1: https://www.nature.com/articles/s41587-025-02777-8.pdf Controller-registered target occurrence locator(s): occ-0001 (document_7f2e411f81d7a4258a23.txt:p-0003); occ-0002 (document_7f2e411f81d7a4258a23.txt:p-0003); occ-0003 (document_7f2e411f81d7a4258a23.txt:p-0004); occ-0004 (document_7f2e411f81d7a4258a23.txt:p-0004); occ-0005 (document_7f2e411f81d7a4258a23.txt:p-0007); occ-0006 (document_7f2e411f81d7a4258a23.txt:p-0007); occ-0007 (document_7f2e411f81d7a4258a23.txt:p-0007); occ-0008 (document_7f2e411f81d7a4258a23.txt:p-0008); occ-0009 (document_7f2e411f81d7a4258a23.txt:p-0008); occ-0010 (document_7f2e411f81d7a4258a23.txt:p-0008); occ-0011 (document_7f2e411f81d7a4258a23.txt:p-0009); occ-0012 (document_7f2e411f81d7a4258a23.txt:p-0010); occ-0013 (document_7f2e411f81d7a4258a23.txt:p-0012); occ-0014 (document_7f2e411f81d7a4258a23.txt:p-0012); occ-0015 (document_7f2e411f81d7a4258a23.txt:p-0012); occ-0016 (document_7f2e411f81d7a4258a23.txt:p-0012); occ-0017 (document_7f2e411f81d7a4258a23.txt:p-0013); occ-0018 (document_7f2e411f81d7a4258a23.txt:p-0013); occ-0019 (document_7f2e411f81d7a4258a23.txt:p-0013); occ-0020 (document_7f2e411f81d7a4258a23.txt:p-0013); occ-0021 (document_7f2e411f81d7a4258a23.txt:p-0014)	Review status:LOCKED Citation validity:Yes Method executed:Yes Uses (select all): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: We considered three state-of-the-art perturbation response prediction methods: compositional perturbation autoencoder (CPA)10, GEARS1 and scGPT2 (Methods). Notes:
GEARS-H V-037 FORMAL VALIDATION	Perturbomics: CRISPR–Cas screening-based functional genomics approach for drug target discovery DOI: 10.1038/s12276-025-01487-0 Article page: https://doi.org/10.1038/s12276-025-01487-0 Full text 1: https://www.nature.com/articles/s12276-025-01487-0.pdf Controller-registered target occurrence locator(s): occ-0001 (document_77500ed58c4fa6921e2d.txt:p-0010); occ-0002 (document_77500ed58c4fa6921e2d.txt:p-0010); occ-0003 (document_77500ed58c4fa6921e2d.txt:p-0012)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient

		Supporting quote + locator: Leskovec and colleagues, for instance, developed the graph-enhanced gene activation and repression simulator (GEARS), which integrates CRISPRa screening data with previously published Perturb-seq datasets to predict combinatorial perturbation effects on gene interactions¹⁰⁰. Notes:
GEARS-H V-021 FORMAL VALIDATI ON	Multifaceted Representation of Genes via Deep Learning of Gene Expression Networks DOI: 10.1101/2024.03.07.583777 Article page: https://doi.org/10.1101/2024.03.07.583777 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2024/03/09/2024.03.07.583777.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_ce413a5d2d0e6225b875.txt:p-0013)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Recent studies have employed advanced AI techniques to model the complexities of gene networks using large gene expression datasets^{1–11}

FORMAL VALIDATI ON	DOI: 10.1101/2023.10.16.562592 Article page: https://doi.org/10.1101/2023.10.16.562592 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/10/19/2023.10.16.562592.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_d96dc927725a902640fe.txt:p-0115)	Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: As more Perturb-seq datasets, as the one generated here, become available across a more diverse range of cell types and conditions, it will be possible to use Perturb-seq data more effectively to extrapolate from one context to another with increasing accuracy^{70,71} and, in the case of malignant cell states, using CNA-to-RNA and CNA-to-cell-state associations to further guide Perturb-seq experimental design. Notes:
GEARS-H V-007 FORMAL VALIDATI ON	Impact of the Human Cell Atlas on medicine DOI: 10.1038/s41591-022-02104-7 Article page: https://doi.org/10.1038/s41591-022-02104-7 Full text 1: https://www.nature.com/articles/s41591-022-02104-7.pdf Controller-registered target occurrence locator(s): occ-0001 (document_80ad0abf109b97c993e8.txt:p-1226)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only

		Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Moreover, machine-learning algorithms can increasingly be trained on such data to yield models that predict the impact of additional perturbations in one or more genes in the same cellular context or of the same perturbations in new biological contexts^{114,115,116}. Notes:
GEARS-H V-034 FORMAL VALIDATION	Knowledge graph-aided Bayesian active learning for top-K genetic interaction discovery DOI: 10.1038/s41598-025-13972-7 Article page: https://doi.org/10.1038/s41598-025-13972-7 Full text 1: https://www.nature.com/articles/s41598-025-13972-7.pdf Controller-registered target occurrence locator(s): occ-0001 (document_01075fd12409c5a74d4f.txt:p-0033)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: For gene knockdown experiments, active learning strategies have been investigated in several studies^{3,4,5,6,7,8}.

		Notes:
GEARS-H V-019 FORMAL VALIDATION	Knowledge-augmented Graph Machine Learning for Drug Discovery: A Survey DOI: 10.1145/3744237 Article page: https://doi.org/10.1145/3744237 Full text 1: https://pure.au.dk/ws/files/469052743/Knowledge-augmented_Graph_Machine_Learning_for_Drug_Discovery.pdf Controller-registered target occurrence locator(s): occ-0001 (document_ba89fde9a7c4cd903d44.txt:p-0015); occ-0002 (document_ba89fde9a7c4cd903d44.txt:p-0034)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: GEARS [164] Nat. Biotechnol 2024 PredMol ✓ Notes:
GEARS-H V-048 FORMAL VALIDATION	CellPLM: Pre-training of Cell Language Model Beyond Single Cells DOI: 10.1101/2023.10.03.560734 Article page: https://doi.org/10.1101/2023.10.03.560734 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/10/05/2023.10.03.560734.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_fd1cf2273fcbecd90247.txt:p-0016); occ-0002 (document_fd1cf2273fcbecd90247.txt:p-0025); occ-0003 (document_fd1cf2273fcbecd90247.txt:p-0025); occ-0004 (document_fd1cf2273fcbecd90247.txt:p-0025); occ-0005 (document_fd1cf2273fcbecd90247.txt:p-0025); occ-0006 (document_fd1cf2273fcbecd90247.txt:p-0025); occ-0007 (document_fd1cf2273fcbecd90247.txt:p-0025); occ-0008 (document_fd1cf2273fcbecd90247.txt:p-0026); occ-0009 (document_fd1cf2273fcbecd90247.txt:p-0028)	Review status:LOCKED Citation validity:Yes Method executed:Yes Uses (select all): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: We compare the performance between CellPLM and two baselines, i.e., a recent preprint GEARS method

		(Roohani et al., 2022), and scGen (Lotfollahi et al., 2019). Notes:
GEARS-H V-009 FORMAL VALIDATI ON	Unagi: Deep Generative Model for Deciphering Cellular Dynamics and In-Silico Drug Discovery in Complex Diseases DOI: 10.21203/rs.3.rs-3676579/v1 Article page: https://doi.org/10.21203/rs.3.rs-3676579/v1 Full text 1: https://www.researchsquare.com/article/rs-3676579/latest.pdf Controller-registered target occurrence locator(s): occ-0001 (document_f5992f36e601995af4bb.txt:p-0004); occ-0002 (document_f5992f36e601995af4bb.txt:p-0032)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Although GEARS25 and scGen23 are able to perform in-silico perturbations, they weren't designed to process time-series data and require the in-vitro screening of perturbation response of disease and drug treatments as the supervision. Notes:
GEARS-H V-011 FORMAL VALIDATI ON	scEMB: Learning context representation of genes based on large-scale single-cell transcriptomics DOI: 10.1101/2024.09.24.614685 Article page: https://doi.org/10.1101/2024.09.24.614685	Review status:LOCKED Citation validity:Yes Method executed:No

	Full text 1: https://www.biorxiv.org/content/biorxiv/early/2024/09/26/2024.09.24.614685.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_084a1ab5ec97944eb131.txt:p-0020)	Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Moreover, these embedding-based methods¹¹ [Autoencoder¹², VAE^{13–16}, GNN¹⁷], has been widely used to along with the cellular perturbation tasks along with single-cell CRISPR data, to compress the transcriptomics data (mainly single-cell) into the latent space and compare the effects of different perturbations and possess a more accurate molecular measurement as well as in silico-induced perturbations^{18,19}. Notes:
GEARS-H V-010 FORMAL VALIDATION	On the use of variational autoencoders for biomedical data integration DOI: 10.1101/2025.08.18.670835 Article page: https://doi.org/10.1101/2025.08.18.670835 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/08/22/2025.08.18.670835.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_136c9cb018fe55dbb9ba.txt:p-0139)	Review status:LOCKED Citation validity:Yes Method executed:Yes Uses (select all): Primary use:Benchmark Biological insight:No

		Evidence sufficiency:Sufficient Supporting quote + locator: These values were comparable to those of the other models benchmarked in a single gene KD setting by Ahlmann-Eltze et al., including foundation models such as scGPT (L2RPE1=6.88, L2K562=4.41) graph-based models like GEARS (L2RPE1=6.33, L2K562=4.04), and their proposed bilinear baseline (L2RPE1=5.84, L2K562=4.04) (Figure 6a,b, Supplementary Figure 16). Notes:
GEARS-H V-026 FORMAL VALIDATION	HyperdCas12a-based multiplexed genetic regulation in Candida albicans DOI: 10.1093/nar/gkaf1402 Article page: https://doi.org/10.1093/nar/gkaf1402 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/10/20/2025.10.20.683421.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_5d55b450a4b894ee372a.txt:p-0094)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Recent technological

		advances in model systems have enabled the identification of a myriad of genetic interactions and non-additive phenotypic outcomes caused by changes in two or more genes [17–21]. Notes:
GEARS-H V-001 FORMAL VALIDATION	Modeling interpretable correspondence between cell state and perturbation response with CellCap DOI: 10.1101/2024.03.14.585078 Article page: https://doi.org/10.1101/2024.03.14.585078 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2024/03/16/2024.03.14.585078.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_445c5329884eb9425d2f.txt:p-0003); occ-0002 (document_445c5329884eb9425d2f.txt:p-0003); occ-0003 (document_445c5329884eb9425d2f.txt:p-0011); occ-0004 (document_445c5329884eb9425d2f.txt:p-0015); occ-0005 (document_445c5329884eb9425d2f.txt:p-0015); occ-0006 (document_445c5329884eb9425d2f.txt:p-0024)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: or example, scGen uses a deep generative model to predict 67 the impact of one perturbation on a new cell population [10], and GEARS takes advantage of 68 prior biological knowledge to model nonlinear gene interactions and nonlinearities in response 69 to multiple perturbations [11].

		Notes:
GEARS-H V-015 FORMAL VALIDATION	MultiPert: An adversarial alignment and dual attention framework for single-cell multi-omics perturbation prediction DOI: 10.1371/journal.pcbi.1014054 Article page: https://doi.org/10.1371/journal.pcbi.1014054 Full text 1: https://journals.plos.org/ploscompbiol/article/file?id=10.1371/journal.pcbi.1014054&type=printable Controller-registered target occurrence locator(s): occ-0001 (document_9719522b17e400f49ce6.txt:p-0003); occ-0002 (document_9719522b17e400f49ce6.txt:p-0004); occ-0003 (document_9719522b17e400f49ce6.txt:p-0010); occ-0004 (document_9719522b17e400f49ce6.txt:p-0020); occ-0005 (document_9719522b17e400f49ce6.txt:p-0020); occ-0006 (document_9719522b17e400f49ce6.txt:p-0020); occ-0007 (document_9719522b17e400f49ce6.txt:p-0020); occ-0008 (document_9719522b17e400f49ce6.txt:p-0020); occ-0009 (document_9719522b17e400f49ce6.txt:p-0021); occ-0010 (document_9719522b17e400f49ce6.txt:p-0021); occ-0011 (document_9719522b17e400f49ce6.txt:p-0023)	Review status:LOCKED Citation validity:Yes Method executed:Yes Uses (select all): Primary use:Extend Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Given GEARS' prior validation of GO knowledge graph for perturbation prediction, we directly adopted their pre-trained embeddings to initialize MultiPert's perturbation representations [16]. Notes:
GEARS-H V-003 FORMAL VALIDATION	rbio1 - training scientific reasoning LLMs with biological world models as soft verifiers DOI: 10.1101/2025.08.18.670981 Article page: https://doi.org/10.1101/2025.08.18.670981 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/08/21/2025.08.18.670981.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_de87e74d675a63fc9796.txt:p-0013); occ-0002 (document_de87e74d675a63fc9796.txt:p-0014); occ-0003	Review status:LOCKED Citation validity:Yes Method executed:Yes Uses (select all): Primary use:Benchmark Biological insight:No

	(document_de87e74d675a63fc9796.txt:p-0014); occ-0004 (document_de87e74d675a63fc9796.txt:p-0026)	Evidence sufficiency:Sufficient Supporting quote + locator: We compare against: SUMMER [32] (retrieval + knowledge-based reasoning), GEARS [38] (a specialized perturbation model), and the base Qwen2.5-3B reasoning model without biological supervision. Notes:
GEARS-H V-018 FORMAL VALIDATION	xTrimoGene: An Efficient and Scalable Representation Learner for Single-Cell RNA-Seq Data DOI: 10.1101/2023.03.24.534055 Article page: https://doi.org/10.1101/2023.03.24.534055 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/03/25/2023.03.24.534055.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_5c94df88575250ab20d9.txt:p-0012); occ-0002 (document_5c94df88575250ab20d9.txt:p-0012); occ-0003 (document_5c94df88575250ab20d9.txt:p-0012); occ-0004 (document_5c94df88575250ab20d9.txt:p-0012); occ-0005 (document_5c94df88575250ab20d9.txt:p-0015); occ-0006 (document_5c94df88575250ab20d9.txt:p-0017); occ-0007 (document_5c94df88575250ab20d9.txt:p-0017); occ-0008 (document_5c94df88575250ab20d9.txt:p-0017); occ-0009 (document_5c94df88575250ab20d9.txt:p-0017)	Review status:LOCKED Citation validity:Yes Method executed:Yes Uses (select all): Primary use:Extend/Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: While utilizing xTrimoGene, the normal state (before perturbation) gene expression profile is fed into xTrimoGene and we obtained the context embedding, which replaces raw expression value input in GEARS model (refer to

		Appendix 7.4 for detailed description). Notes:
GEARS-H V-013 FORMAL VALIDATION	A systematic comparison of computational methods for expression forecasting DOI: 10.1101/2023.07.28.551039 Article page: https://doi.org/10.1101/2023.07.28.551039 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/07/31/2023.07.28.551039.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_02100be951234bc62ab3.txt:p-0005); occ-0002 (document_02100be951234bc62ab3.txt:p-0006); occ-0003 (document_02100be951234bc62ab3.txt:p-0018); occ-0004 (document_02100be951234bc62ab3.txt:p-0020); occ-0005 (document_02100be951234bc62ab3.txt:p-0020); occ-0006 (document_02100be951234bc62ab3.txt:p-0021); occ-0007 (document_02100be951234bc62ab3.txt:p-0021); occ-0008 (document_02100be951234bc62ab3.txt:p-0021); occ-0009 (document_02100be951234bc62ab3.txt:p-0021); occ-0010 (document_02100be951234bc62ab3.txt:p-0021); occ-0011 (document_02100be951234bc62ab3.txt:p-0021); occ-0012 (document_02100be951234bc62ab3.txt:p-0021); occ-0013 (document_02100be951234bc62ab3.txt:p-0021); occ-0014 (document_02100be951234bc62ab3.txt:p-0021); occ-0015 (document_02100be951234bc62ab3.txt:p-0022); occ-0016 (document_02100be951234bc62ab3.txt:p-0022); occ-0017 (document_02100be951234bc62ab3.txt:p-0022); occ-0018 (document_02100be951234bc62ab3.txt:p-0023); occ-0019 (document_02100be951234bc62ab3.txt:p-0023); occ-0020 (document_02100be951234bc62ab3.txt:p-0024); occ-0021 (document_02100be951234bc62ab3.txt:p-0025); occ-0022 (document_02100be951234bc62ab3.txt:p-0025); occ-0023 (document_02100be951234bc62ab3.txt:p-0027); occ-0024 (document_02100be951234bc62ab3.txt:p-0027); occ-0025 (document_02100be951234bc62ab3.txt:p-0027); occ-0026 (document_02100be951234bc62ab3.txt:p-0027); occ-0027 (document_02100be951234bc62ab3.txt:p-0033); occ-0028 (document_02100be951234bc62ab3.txt:p-0039); occ-0029 (document_02100be951234bc62ab3.txt:p-0039); occ-0030 (document_02100be951234bc62ab3.txt:p-0039); occ-0031 (document_02100be951234bc62ab3.txt:p-0039)	Review status:LOCKED Citation validity:Yes Method executed:Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: We included all methods with peer-reviewed paper and code available as of Sept 1, 2023, which were DCD-FG, GEARS, and GeneFormer (Lopez et al., 2022; Roohani et al., 2022; Theodoris et al., 2023). Notes:
GEARS-H V-022	Learning Cross-Domain Representations for Transferable Drug Perturbations on Single-Cell Transcriptional Responses	Review status:LOCKED

FORMAL VALIDATION	DOI: 10.1609/aaai.v39i18.34073 Article page: https://doi.org/10.1609/aaai.v39i18.34073 Full text 1: https://ojs.aaai.org/index.php/AAAI/article/download/34073/36228 Controller-registered target occurrence locator(s): occ-0001 (document_5ee8066e213823971bab.txt:p-0002); occ-0002 (document_5ee8066e213823971bab.txt:p-0009); occ-0003 (document_5ee8066e213823971bab.txt:p-0010); occ-0004 (document_5ee8066e213823971bab.txt:p-0010); occ-0005 (document_5ee8066e213823971bab.txt:p-0010); occ-0006 (document_5ee8066e213823971bab.txt:p-0010); occ-0007 (document_5ee8066e213823971bab.txt:p-0012)	Citation validity:Yes Method executed:Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Additionally, to evaluate our model's performance, we compared it against previously published methods, including chemCPA (Hetzel et al. 2022), GEARS (Roohani, Huang, and Leskovec 2022) and cycleCDR (Huang and Liu 2024). Notes:
GEARS-H V-005 FORMAL VALIDATION	GexMolGen: cross-modal generation of hit-like molecules via large language model encoding of gene expression signatures DOI: 10.1093/bib/bbae525 Article page: https://doi.org/10.1093/bib/bbae525 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/11/26/2023.11.11.566725.full.pdf No controller-registered target occurrence; independently audit citation location and evidence coverage.	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Extend Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Instead of commonly using

		mean square loss, we use the autofocus direction-aware loss in GEARS [32]. Notes:
GEARS-H V-042 FORMAL VALIDATION	Harnessing the power of single-cell large language models with parameter-efficient fine-tuning using scPEFT DOI: 10.1038/s42256-025-01170-z Article page: https://doi.org/10.1038/s42256-025-01170-z Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/04/23/2025.04.21.649754.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_186c573a618178144ab2.txt:p-0030); occ-0002 (document_186c573a618178144ab2.txt:p-0030); occ-0003 (document_186c573a618178144ab2.txt:p-0034); occ-0004 (document_186c573a618178144ab2.txt:p-0066); occ-0005 (document_186c573a618178144ab2.txt:p-0066); occ-0006 (document_186c573a618178144ab2.txt:p-0070); occ-0007 (document_186c573a618178144ab2.txt:p-0070); occ-0008 (document_186c573a618178144ab2.txt:p-0070); occ-0009 (document_186c573a618178144ab2.txt:p-0070); occ-0010 (document_186c573a618178144ab2.txt:p-0072); occ-0011 (document_186c573a618178144ab2.txt:p-0072); occ-0012 (document_186c573a618178144ab2.txt:p-0072); occ-0013 (document_186c573a618178144ab2.txt:p-0072); occ-0014 (document_186c573a618178144ab2.txt:p-0100); occ-0015 (document_186c573a618178144ab2.txt:p-0100)	Review status:LOCKED Citation validity:Yes Method executed:Yes Uses (select all): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: It improved the mean squared error (MSE) and Pearson's correlation coefficient (PCC) for the top 20 most differentially expressed genes (MSE_DE and PCC_DE) by 0.06 and up to 0.15, respectively, compared with zero-shot scFoundation, and by 0.02–0.05 and up to 0.04, respectively, compared with GEARS29 (Supplementary Table 5). Notes:

GEARS-H V-030 FORMAL VALIDATION	Imputing spatial transcriptomics through gene network constructed from protein language model DOI: 10.1038/s42003-024-06964-2 Article page: https://doi.org/10.1038/s42003-024-06964-2 Full text 1: https://www.nature.com/articles/s42003-024-06964-2.pdf Controller-registered target occurrence locator(s): occ-0001 (document_3d5b0c1d4207d32a4481.txt:p-0011)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Most importantly, TransImpLR ignores gene-to-gene relationships in gene prediction, which have shown improved performance in previous work^{15,16}. Notes:
GEARS-H V-033 FORMAL VALIDATION	Molecular causality in the advent of foundation models DOI: 10.1038/s44320-024-00041-w Article page: https://doi.org/10.1038/s44320-024-00041-w Full text 1: https://link.springer.com/content/pdf/10.1038/s44320-024-00041-w.pdf Controller-registered target occurrence locator(s): occ-0001 (document_43a44b24877ae33f7710.txt:p-0010)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Hybrid models make a tradeoff between those extremes,

		which is why they have been found to be useful in systems biology, where data are currently scarce (AlQuraishi and Sorger, 2021; Nilsson et al, 2022; Faure et al, 2023; Roohani et al, 2023; Fortelny and Bock, 2020; Lotfollahi et al, 2023; Yuan et al, 2021). Notes:
GEARS-H V-020 FORMAL VALIDATI ON	Fluctuation structure predicts genome-wide perturbation outcomes DOI: 10.1101/2025.06.27.661814 Article page: https://doi.org/10.1101/2025.06.27.661814 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/07/01/2025.06.27.661814.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_726b71ecff5ac6626a97.txt:p-0007); occ-0002 (document_726b71ecff5ac6626a97.txt:p-0082)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: At the same time, traditional linear regression models have some limitations: they typically treat genes as independent, cannot explicitly identify the genes driving the response (or the degree of impact per driver gene), and, with the exception of TRADE16, produce only point estimates of the regression coefficients and are

		not uncertainty-aware. Predictive models such as scGen20, GEARS21, and scGPT22, on the other hand, employ deep learning frameworks to model perturbation responses in latent spaces and can predict outcomes of unseen perturbations. Notes:
GEARS-H V-039 FORMAL VALIDATION	Cell Decoder: decoding cell identity with multi-scale explainable deep learning DOI: 10.1186/s13059-025-03832-y Article page: https://doi.org/10.1186/s13059-025-03832-y Full text 1: https://genomebiology.biomedcentral.com/counter/pdf/10.1186/s13059-025-03832-y Controller-registered target occurrence locator(s): occ-0001 (document_86b5e3affeed65eab298.txt:p-0029)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Due to the incorporation of gene/protein interaction information, Cell Decoder opens up the possibility of predicting cell responses to genetic perturbations [42]. Notes:

GEARS-H V-012 FORMAL VALIDATION	Evaluating the Utilities of Large Language Models in Single-cell Data Analysis DOI: 10.21203/rs.3.rs-3376641/v1 Article page: https://doi.org/10.21203/rs.3.rs-3376641/v1 Full text 1: https://www.researchsquare.com/article/rs-3376641/latest.pdf Controller-registered target occurrence locator(s): occ-0001 (document_94ec69cb17831b26e059.txt:p-0009); occ-0002 (document_94ec69cb17831b26e059.txt:p-0010); occ-0003 (document_94ec69cb17831b26e059.txt:p-0010); occ-0004 (document_94ec69cb17831b26e059.txt:p-0011); occ-0005 (document_94ec69cb17831b26e059.txt:p-0024); occ-0006 (document_94ec69cb17831b26e059.txt:p-0024); occ-0007 (document_94ec69cb17831b26e059.txt:p-0031)	Review status:LOCKED Citation validity:Yes Method executed:Yes Uses (select all): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: We considered scGPT and GEARS [42] for this task. In the training process, we masked the genes under perturbation and tried to reconstruct the expression levels of all the genes of the input cells rather than only the masked genes. Notes:
GEARS-H V-038 FORMAL VALIDATION	scYeast: a Biological-knowledge-guided Foundation Model on Yeast Single-Cell Transcriptomics DOI: 10.1101/2025.08.20.671179 Article page: https://doi.org/10.1101/2025.08.20.671179 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/08/25/2025.08.20.671179.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_7216842b65d3d2e24031.txt:p-0025); occ-0002 (document_7216842b65d3d2e24031.txt:p-0067)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Extend Biological insight:No Evidence sufficiency:Sufficient

		Supporting quote + locator: Since scYeast has already integrated prior knowledge such as gene co-expression and transcriptional regulatory relationships during its pre-training, we adapted the GEARS [32] framework by using scYeast as the base model for predicting gene perturbation responses (Methods) (Fig. 4a). Notes:
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